

MARSS: Radar Semantic Segmentation via Modular Attention and State Space Models

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Abstract

Radar semantic segmentation (RSS) is critical for robust perception in adverse conditions, but poses unique challenges: radar frequency maps are highly anisotropic, multi-scale, sparse and noisy. Conventional CNN or Transformer architectures, designed for camera images, fail to account for these characteristics, leading degraded performance. We propose MARSS (Modular Attention-enhanced Radar Semantic Segmentation), a novel framework that integrates three specialized modules to address radar-specific issues. In the encoder, the RADE module employs lightweight channel self-attention and depthwise convolutions to robustly encode noisy, anisotropic features. In intermediate layers, the RFAF module performs multi-scale feature fusion and region-level attention to isolate salient radar features. The decoder's RADM module combines state space models with axial self-attention to reconstruct segmentation masks with anisotropy and temporality-aware context. These components collectively suppress noise, disentangle range-Doppler features, and enforce spatial-temporal consistency. On the CARRADA dataset, MARSS achieves substantially higher performance than prior RSS methods, especially for small fast-moving targets.

1. Introduction

Radar sensors are increasingly used in autonomous driving[23] and robotics[13] for their robustness in low-visibility or adverse weather. Unlike RGB images, radar returns are typically represented as frequency-domain maps

(e.g., Range-Doppler and Range-Angle). This frequency map provides rich information about object velocities, but also poses unique challenges for semantic segmentation.

First, radar frequency maps are anisotropic[18], with each axis representing different physical quantities. Patterns in radar images exhibit strong directional bias: features behave differently along the range and velocity dimensions, as shown in Figure 1. This contrasts with the isotropic texture patterns in RGB imagery[20]. Traditional CNNs, designed for camera images, are not well-suited to capture this horizontal/vertical contextual imbalance.

Second, radar imaging exhibits inherent spatial multi-scale characteristics. For instance, a large, slowly moving object generates broad, low-frequency reflections (spread across range bins but narrow in Doppler), whereas a small, fast-moving object produces sharp high-frequency streaks (narrow in range but extended in Doppler). Segmentation models must move beyond fixed receptive fields to handle such multi-scale frequency content[29].

Besides, the radar image is extremely sparse and noisy[15]. Most of the pixels in the radar Fast Fourier Transform (FFT) image exhibit almost no echoes (black background), with sparse peaks appearing only at the locations of actual reflectors such as vehicles and pedestrians. Additionally, the radar signal is affected by speckle noise and multipath artifacts, leading to false responses in certain regions and a low signal-to-noise ratio. Small targets may be only slightly above the noise floor. This high dynamic range and noise make it difficult for segmentation networks to learn effective features.

To address these limitations, we propose MARSS, a Modular Attention-enhanced Radar Semantic Segmentation framework that embeds radar-specific design princi-

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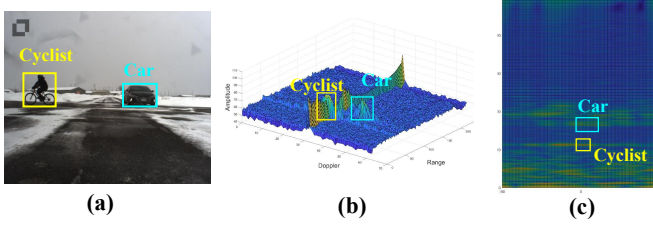


Figure 1. From left to right, figures illustrate (a) the synchronized camera image, (b) the Range-Doppler (RD) input in 3D Range-Doppler-Amplitude representation, and (c) the 2D Range-Angle (RA) representation. Radar frequency maps are inherently three-dimensional, spanning the range, angle and Doppler axes. Projecting along any one dimension yields a corresponding perspective view. As shown in the figure, the radar frequency data exhibits strong anisotropy across different directions, with targets appearing as sharp peaks embedded in substantial background noise.

ples into each stage of the network. MARSS comprises three novel modules inspired by recent advances: in the encoder, RADE uses channel and spatial attention[24] to robustly encode noisy features; at feature fusion layers, RFAF provides multi-scale feature fusion and region-level attention[21]; in the decoder, RADM combines Mamba state space models[5, 12] with axial self-attention to effectively handle the anisotropy and long-range dependencies in radar data. These modules complement each other to address radar’s low SNR, anisotropy, and temporal dynamics.

We evaluate MARSS on the public CARRADA dataset[17]. Results show that MARSS significantly outperforms prior methods, particularly boosting robustness on small and fast targets detection, ablation studies confirm each module’s benefit and complementarity. In summary, our contributions are:

- MARSS framework: We introduce a modular RSS network that embeds radar-specific inductive biases (anisotropy, sparsity, multi-scale) via following modules:
 - a) RADE module: A new encoder block combining channel attention and spatial attention mechanisms to suppress noise and recalibrate channels.
 - b) RFAF module: Radar Feature Adaptive Fusion module that isolates salient features across scales and locations in radar frequency maps.
 - c) RADM decoder: A Mamba-based Decoder combining Mamba state space models with axial self-attention to efficiently handle anisotropy and long-range dependencies in radar data.
- Our results demonstrate the effectiveness of modular design for radar perception, setting new state-of-the-art on CARRADA and CARRADA-RAC datasets[17, 26].

2. Related Work

2.1. Radar Semantic Segmentation

RSS methods have advanced considerably through the adaptation of techniques from other sensing modalities and the development of radar-specific operators. Early approaches primarily transfer classical point cloud-based[23] or image-based processing pipelines to radar data. For instance, ERASE-Net projects radar returns into point-cloud representations processed by sparse 3D CNNs, while RSS-Net[8] adopts a U-Net[19] variant for frequency map segmentation. RAMP-CNN[4] further introduces multi-scale convolutions and spatio-temporal blocks to capture motion cues. More recent Transformer-based models (TransRSS[30]) and multi-view fusion approaches (TransRadar[2]) improve global context modeling and cross-view consistency across Range-Doppler and Range-Angle domains. In parallel, radar-tailored operators such as PeakConv[26], AdaPKC[11] and TARSS-Net[28] emphasize peak responses or temporal coherence. Despite these advances, most existing methods do not explicitly disentangle the directional components embedded in the radar spectrum. In contrast, our work directly targets this limitation and introduces attention modules designed to adaptively isolate directional and multi-scale information.

2.2. Image Semantic Segmentation

In image semantic segmentation, encoder–decoder architectures such as U-Net [19] remain dominant paradigms. To enhance contextual perception across multiple scales, the Atrous Spatial Pyramid Pooling (ASPP) module [1] has been widely adopted for feature enhancement. Meanwhile, various attention mechanisms have further improved feature refinement, with SENet [7] modeling inter-channel dependencies and CBAM [24] combining channel and spatial attention. Inspired by these designs, our proposed RFAF module integrates multi-scale feature fusion with both channel and spatial attention to accommodate the sparse and noisy characteristics of radar signals. Beyond pixel-level attention, region-level attention [21] has been shown effective in highlighting task-relevant regions. Motivated by this, we adapt a similar region-level attention scheme to radar representation learning, emphasizing radar returns corresponding to physical targets while suppressing clutter. To further capture long-range dependencies, axial self-attention [22] decomposes 2D attention along individual axes, while structured state-space models (SSMs) [5] offer strong sequential modeling capability. Our RADM module integrates SSM-based modeling with axial attention to separately capture dependencies along the range and Doppler axes, effectively combining the efficiency of state-space dynamics with the expressiveness of attention mechanisms. Overall, the proposed modules introduce radar-specific priors, into the radar

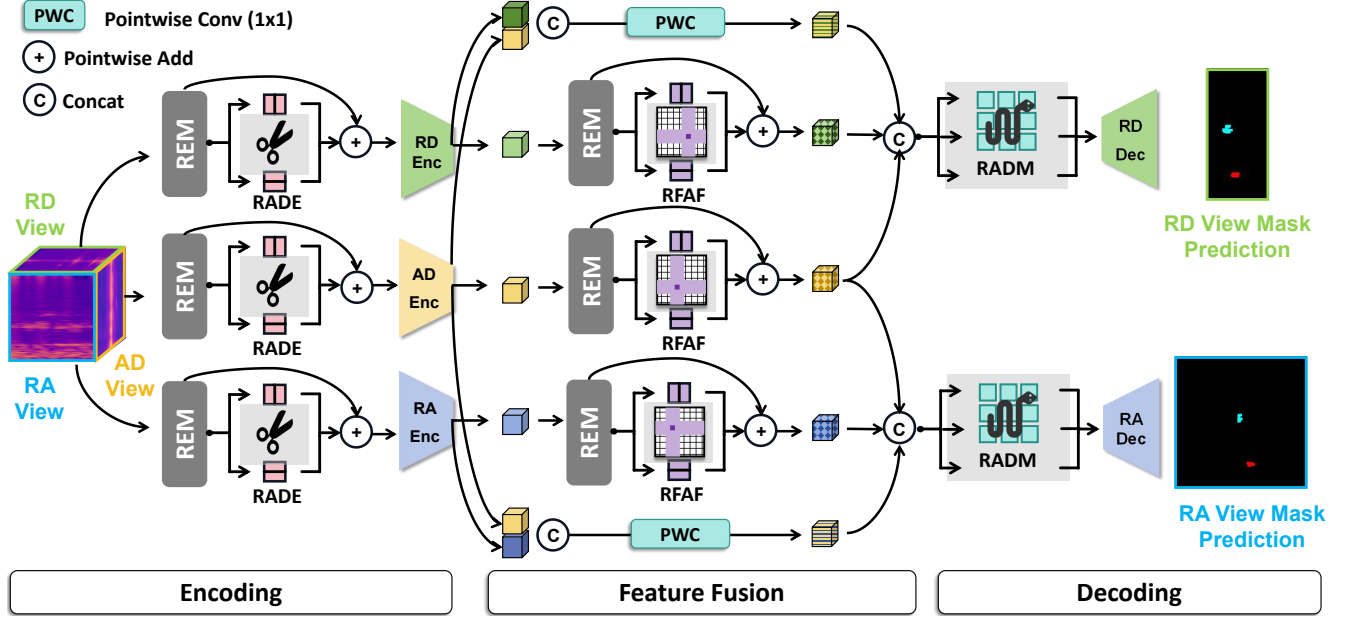


Figure 2. The Proposed MARSS Network Architecture: MARSS takes three perspective views of the radar frequency maps as input. Each view is first enhanced by the RADE module to refine original features and then passed through encoder branches. Intermediate features are further strengthened by the RFAF module and concatenated with their unenhanced counterparts. These are subsequently reconstructed via RADM and decoded into semantic segmentation masks using a decoder.

semantic segmentation task, thereby bridging the gap between visual architectures and radar perception models.

3. Method

The overall MARSS architecture follows an encoder-decoder paradigm[16] with specialized modules at each stage (as shown in Figure.2). We process the three perspective images of the radar frequency map (range-Doppler, angle-Doppler and range-angle) via separate paths and decode them into range-Doppler and range-angle object masks at the output. Our proposed three radar feature enhancement modules (RADE, RFAF and RADM) are embedded in the encoder, intermediate feature interaction, and decoder stages of the segmentation pipeline to provide strong radar perception guidance. We also adopted a wavelet transform guided REM (Radar Enhancement Module) in the encoding stage and the feature fusion stage for axial decoupling and enhancement of anisotropic radar features.

3.1. REM: Radar Enhancement Module

The REM is based on wavelet transform for the encoding stage and intermediate feature fusion stage of the pipeline. The 2D discrete wavelet transform decouples the signal into subbands LL, LH, HL and HH which contains low-frequency, high-frequency and diagonal components. The 2D DWT enabling multi-resolution and multi-orientation analysis[3, 14]. Each subsequent branch will focus on a

specific subband, facilitating the learning of anisotropic and multi-band characteristics in radar frequency maps.

As illustrated in Figure 3, each REM unit consists of four branches like wavelet transform: the LL branch employs a 1×1 convolution to provide global low-frequency information, while the HH branch further incorporates a 3×3 convolution to model diagonal high-frequency components. The LH branch first applies a vertical 1×3 axial convolution to capture low-frequency features along the vertical direction, followed by a horizontal 3×1 axial convolution and a standard 3×3 convolution to aggregate horizontal high-frequency features; the HL branch performs the reverse.

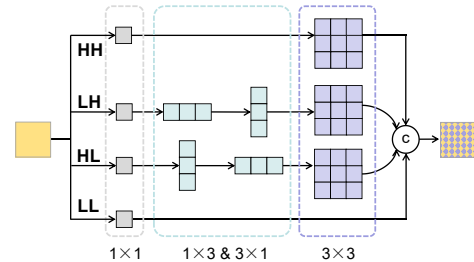


Figure 3. REM: Radar Enhancement Module

3.2. RADE: Radar-Aware Denoising Encoder

Radar spectra are inherently low-SNR and exhibit anisotropic clutter patterns, making both channel distortions

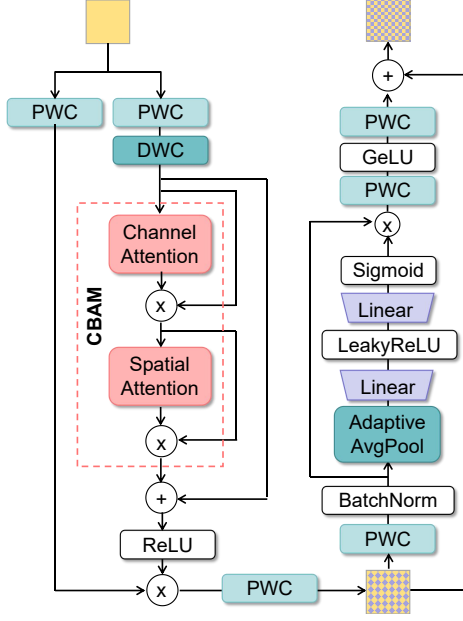


Figure 4. RADE: Radar-Aware Denoising Encoder. Note that DWC represents depth-wise convolution, PWC represents point-wise convolution(e.g., 1x1 convolution).

and spatial interference problematic for high-quality semantic segmentation. To obtain noise-resilient and structure-preserving features, we introduce the Radar-Aware Denoising Encoder (RADE), which integrates lightweight channel calibration with a CBAM-style[24, 27] dual-attention refinement. Inserted after REM feature enhancement, RADE ensures that subsequent layers receive noise-suppressed and target-enhanced features. The enriched features are then fed into the encoder, where we adopt 3D CNN.

Given an input $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, RADE first applies a 1×1 projection with GELU to produce a spectrally compressed representation. A squeeze-and-excitation style channel weighting is used to suppress noise-dominated channels:

$$\mathbf{X}_c = \sigma_{\text{sig}}(f_{\text{MLP}}(\text{GAP}(\mathbf{X}))) \odot \mathbf{X}, \quad (1)$$

where GAP extracts global spectral statistics and f_{MLP} yields channel importance coefficients. This operation stabilizes the radar frequency bins by down-weighting channels affected by interference or multipath fading.

To further highlight geometrically meaningful regions in the range-Doppler grid, RADE applies a spatial attention mask computed from channel-aggregated descriptors:

$$\mathbf{X}_s = \sigma_{\text{sig}}(\text{Conv}_{7 \times 7}([\text{AvgPool}; \text{MaxPool}](\mathbf{X}_c))) \odot \mathbf{X}_c. \quad (2)$$

This spatial mask enhances local reflectivity structures such as object boundaries or micro-motion streaks, which are essential cues for semantic segmentation in radar imagery.

By chaining channel denoising and spatial structure modeling, RADE performs view-specific noise suppression and geometry-aware enhancement simultaneously. The channel attention mitigates instability across radar receivers or frequency bins, while the spatial attention emphasizes coherent target patterns despite clutter. As a result, RADE provides a robust, low-noise latent representation, improving segmentation accuracy in challenging radar scenes.

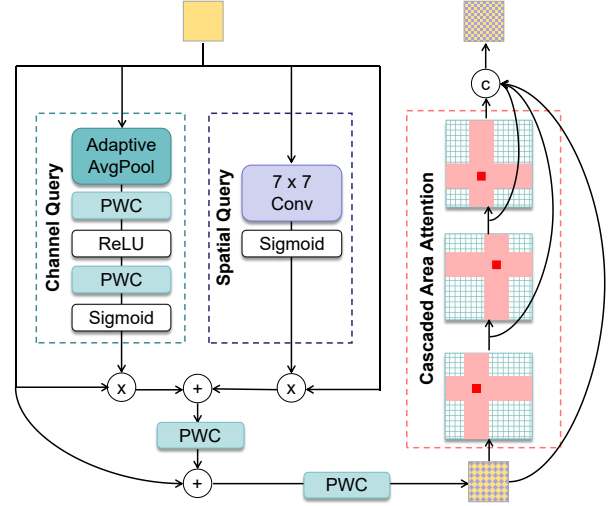


Figure 5. RFAF: Radar Feature Adaptive Fusion Module. PWC represents point-wise convolution (e.g., 1x1 convolution).

3.3. RFAF: Radar Feature Adaptive Fusion Module

In the network’s intermediate layers, radar features often exhibit complex multi-scale patterns, requiring simultaneous processing of features at different scales and attention to important regions. We introduce Radar Feature Adaptive Fusion (RFAF) module, which integrates multi-scale feature fusion and region-level attention, specifically designed to handle the characteristics of radar data.

The RFAF module first extracts features at multiple resolutions via atrous or pooling layers, producing fused feature x_{fused} . This fused feature is then fed into two parallel processing stages:

Multi-scale Fusion Stage[25] feeds x_{fused} into two attention branches: a channel-attention branch and a spatial-attention branch. In the channel-attention branch, we apply Global Average Pooling, a 1×1 bottleneck conv, ReLU, then a 1×1 conv expansion and sigmoid to yield channel weights ch_{attn} . We compute $x_{\text{channel}} = x_{\text{fused}} \odot ch_{\text{attn}}$. In the spatial-attention branch, we apply a 7×7 convolution + sigmoid to produce spatial weights sp_{attn} , yielding $x_{\text{spatial}} = x_{\text{fused}} \odot sp_{\text{attn}}$. These attention-modulated maps are then element-wise added and fused with a 1×1 conv.

Region Attention Stage[21] operates on the feature map (B, C_{in}, H, W) and captures region-specific importance

through a dual-branch design. To further model radar-specific anisotropy, RFAF incorporates a region-attention unit that processes horizontal and vertical neighborhoods separately, forming a cross-shaped receptive pattern. One branch performs region-level attention, while another applies multi-scale dilated operations; the outputs are concatenated and projected back to C channels:

$$\mathbf{Y} = g([\text{RegAtt}(\mathbf{X}_{c+s}) \parallel \text{MS-Dilated}(\mathbf{X}_{c+s})]), \quad (3)$$

where $\mathbf{X}_{c+s}$ is the fused output from Stage I. This attention stacking explicitly captures directional patterns common in radar spectra—horizontal Doppler trails, vertically elongated range responses, and small target blobs—allowing the network to treat each region according to its anisotropy.

Finally, the outputs of the multi-scale fusion branch and region attention branch are fused, with a residual connection adding the original input features. The RFAF module adaptively enhances salient multi-scale features and suppresses noise, separating different frequency bands and spatial patterns in the radar map while highlighting important local regions. This is particularly useful for separating overlapping features in the anisotropic radar spectrum.

3.4. RADM: Radar State Space Decoder Module

The decoder aims to reconstruct the segmentation mask from encoded features while effectively handling the anisotropy and long-range dependencies in radar data. We design a Radar State Space Decoder (RADM) module that combines the advantages of Mamba state space models[5] and axial self-attention. RADM splits the input (B, C, H, W) into three branches that capture different interactions[10]:

1. **Spatial Branch:** This branch reorders the tensor to flatten the width dimension: shape $(B \times W, H, C)$. A linear projection reduces channels to c_1 . We then apply a Vision State Space Block (VSSB) based on Mamba for local spatial-channel interaction, followed by Axial Self-Attention (ASA) along the remaining axis to model long-range dependencies, yielding features of shape $(B \times W, H, c_1)$. These are reshaped back to (B, c_1, H, W) . The VSSB adaptively focuses on important features through selective state space mechanisms while maintaining linear computational complexity.
2. **Doppler Branch:** Here we flatten the height axis $(B \times H, W, C)$ and reduce to c_2 channels. After VSSB local interaction and axial self-attention, we reshape back to (B, c_2, H, W) . This branch explicitly models dependencies along each axis (range vs Doppler) separately, leveraging state space models to effectively capture anisotropic structures.

The outputs of the three branches (channels c_1, c_2, c_1) are concatenated to form $(B, 2c_1 + c_2, H, W)$. A 1×1 conv fuses these into C channels, matching the input depth. A

residual connection adds the original input feature map. The core advantage of the RADM module lies in combining the benefits of state space models and axial attention:

- **State Space Models[12]:** Adaptively focus on important features through selective mechanisms, maintaining linear computational complexity, suitable for handling long-sequence characteristics of radar data.
- **Axial Self-Attention[10]:** Model dependencies along range and Doppler axes separately, effectively handling anisotropy in radar data.
- **Multi-branch Structure:** Capture feature representations from different dimensions, providing complementary contextual information.

This combination enables RADM to handle the complex characteristics of radar data while achieving a good balance between computational efficiency and decoding quality.

Finally, after RADM, we apply standard upsampling and convolutional layers to generate the final segmentation mask for each view. We use softmax activation and train with a combination of weighted cross-entropy and Dice loss, ensuring balanced learning across classes.

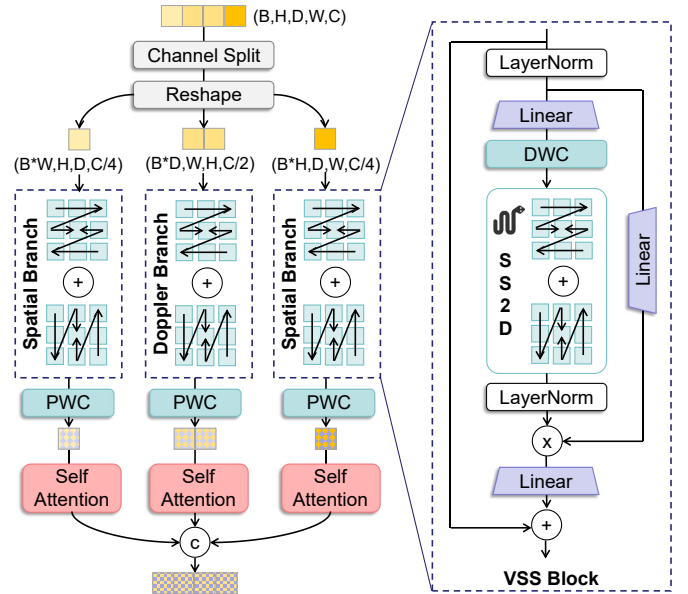


Figure 6. RADM: Radar State Space Decoder Module. DWC represents depth-wise convolution, PWC represents point-wise convolution (e.g., 1×1 convolution).

3.5. Training Loss

After the three-stage encoding-decoding process, MARSS produces semantic segmentation masks from RA and RD views. These are compared with their corresponding ground truth using a multi-task loss. The loss function for each view combines weighted cross-entropy, Dice loss, and temporal coherence loss following [16]:

Table 1. Performance comparison on CARRADA. Results are reported in terms of mIoU and mDice for RD and RA views. Best results are **bold**, second-best are underlined.

Frameworks	RD View		RA View		#Params @ Frames
	mIoU	mDice	mIoU	mDice	
FCN	54.70%	66.30%	34.50%	40.90%	134.3M@3
U-Net	55.40%	68.00%	32.80%	38.20%	17.3M@3
DeepLabv3++	50.80%	61.60%	32.70%	38.30%	59.3M@3
RSS-Net	32.10%	36.90%	32.10%	37.80%	10.1M@3
RAMP-CN	56.60%	68.50%	27.90%	30.50%	106.4M@9
TMVA-Net	56.10%	68.00%	37.70%	46.20%	5.6M@5
TransRadar	57.20%	69.10%	39.90%	49.50%	4.9M@5
T-RODNet	-	-	43.50%	53.60%	162.0M@4
PKCIn-Net	60.70%	72.60%	43.10%	53.70%	6.3M@5
AdaPKC ^ξ -Net	61.20%	73.10%	44.10%	55.10%	6.3M@5
AdaPKC ^θ -Net	61.50%	73.60%	43.60%	54.50%	6.3M@5
AdaPKC ^ξ -Net ^{FiT}	<u>62.10%</u>	<u>74.00%</u>	<u>44.30%</u>	<u>55.50%</u>	6.3M@5
MARSS(ours)	63.26%	75.18%	46.97%	58.78%	9.3M@5

$$\mathcal{L}_{\text{wCE}}(\mathbf{y}, \mathbf{p}) = -\frac{1}{K} \sum_{k=1}^K w_k \sum_{(m,n) \in \Omega} \mathbf{y}[m, n, k] \log \mathbf{p}[m, n, k], \quad (4)$$

$$\mathcal{L}_{\text{SDice}} = \frac{1}{K} \sum_{k=1}^K \left[1 - \frac{2 \sum_{(m,n)} \mathbf{y}[m, n, k] \mathbf{p}[m, n, k]}{\sum_{(m,n)} \mathbf{y}^2[m, n, k] + \mathbf{p}^2[m, n, k]} \right], \quad (5)$$

$$\mathcal{L}_{\text{CoL}}(\mathbf{p}^{RD}, \mathbf{p}^{RA}) = \|\hat{\mathbf{p}}^{RD} - \hat{\mathbf{p}}^{RA}\|_F^2, \quad (6)$$

$$L = \lambda_{\text{wCE}} (\mathcal{L}_{\text{wCE}}^{RD} + \mathcal{L}_{\text{wCE}}^{RA}) + \lambda_{\text{SDice}} (\mathcal{L}_{\text{SDice}}^{RD} + \mathcal{L}_{\text{SDice}}^{RA}) + \lambda_{\text{CoL}} \mathcal{L}_{\text{CoL}}, \quad (7)$$

where $\mathbf{y} \in \{0, 1\}^{M \times N \times K}$ denote the one-hot encoded ground truth tensor, and $\mathbf{p} \in [0, 1]^{M \times N \times K}$ represent the predicted probability tensor, where M and N are the spatial dimensions (height and width) of the radar view, and K is the total number of classes. $\Omega = [1, M] \times [1, N]$ indicates the set of spatial coordinates, with (m, n) being the spatial position indices. w_k denotes the normalized weight for class k , inversely proportional to its frequency in the training set. $\mathbf{p}^{RD}$ and $\mathbf{p}^{RA}$ correspond to the predicted probability maps for the Range-Doppler (RD) and Range-Angle (RA) views respectively. $\hat{\mathbf{p}}^{RD}$ and $\hat{\mathbf{p}}^{RA}$ are the aggregated feature maps obtained by applying the $\max(\cdot)$ operation along the non-shared axis (Doppler or angle dimension). $\|\cdot\|_F$ signifies the Frobenius norm. λ_{wCE} , λ_{SDice} , and λ_{CoL} are weighting coefficients for the respective loss terms.

Each loss component enforces view-specific learning. Training with multi-view segmentation tasks encourages the model to exploit the multi-dimensional structure of radar frequency maps and improves cross-view consistency, thus enhancing segmentation [16].

4. Experiment

4.1. Dataset and Metrics

We evaluate MARSS on the CARRADA [17] and CARRADA-RAC[26] datasets. CARRADA is captured using FMCW millimeter-wave radar, features synchronized camera-radar recordings across multiple scenarios. It contains four object categories: pedestrian, cyclist, car, and background. The dataset provides range-angle-Doppler (RAD) tensors with dimensions $256 \times 256 \times 64$, supporting multi-view RSS through both range-azimuth (RA) and range-Doppler (RD) representations. For all models being compared, the dataset splits follow the same partitioning scheme established in prior works [6, 26] to ensure a strictly fair comparison. CARRADA-RAC is a recalibrated version of the original CARRADA dataset, providing more consistent annotations. We adopt standard metrics: mean Intersection-over-Union (mIoU) and mean Dice coefficient (mDice) on both RD and RA views. Higher mIoU and mDice indicate better segmentation quality.

Implementation Details

MARSS utilize a sequence of five consecutive frames as input for temporal modeling, consistent with the protocol adopted in PeakConv[26]. The input resolutions are set to 256×256 for the Range-Angle (RA) view, and 256×64

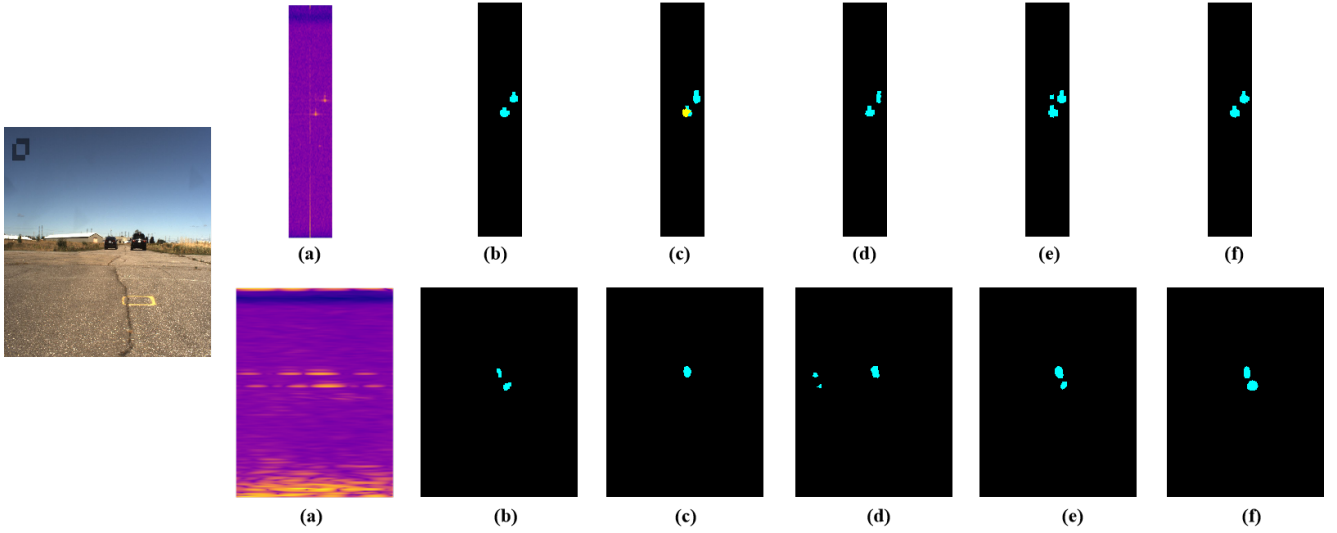


Figure 7. Radar segmentation results visualization: The leftmost image is the corresponding RGB camera image. The first row is RD-View, and the second row is RA-View images. From left to right, they are: (a) original RA/RD data; (b) Segmentation Mask ground truth; (c) TransRadar segmentation results; (d) TMVA-Net segmentation results; (e) AdaPKC^E-Net^{FiT} segmentation results; (f) MARSS(Ours) segmentation results. In the segmentation mask, red corresponds to pedestrians, yellow to cyclists, and blue to cars.

for both the Angle-Doppler (AD) and Range-Doppler (RD) views. Model training is performed on two parallel NVIDIA RTX 3090 GPUs using the Adam optimizer[9]. The initial learning rate is set to 0.0001 and follows an exponential annealing schedule throughout training. Each model is trained for 300 epochs with a batch size of 3. The loss function combines weighted cross-entropy, Dice loss, and temporal coherence loss, using the recommended hyperparameter settings from [6].

4.2. Main Experiments

Our comprehensive evaluation on the CARRADA benchmark demonstrates MARSS’s superior performance against state-of-the-art methods. As shown in Table 1, achieving 63.26% mIoU on RD view and 46.97% mIoU on RA view, directly stems from its principled approach to addressing the fundamental radar challenges outlined in our introduction. The substantial +2.67% mIoU improvement on the challenging RA view over AdaPKC-NetFiT particularly demonstrates MARSS’s effectiveness in handling radar’s inherent anisotropy, where range and Doppler/velocity axes encode fundamentally different physical information. This performance gap is especially significant given that traditional methods struggle with directional feature imbalance in RA representations. The framework’s consistent excellence across both views - also evidenced by its top global performance of 55.2% mIoU on CARRADA-RAC - validates our approach to embedding radar-specific inductive biases throughout the network architecture.

Notably, MARSS achieves these advances with only 9.3M parameters, proving significantly more efficient than methods like T-RODNet (162.0M parameters) while delivering superior results. This efficiency stems from targeted module design: the RADM decoder’s combination of Mamba state space models with axial self-attention directly counteracts the feature mixing problems that affect conventional backbones, enabling adaptive focus on relevant information across long ranges while maintaining linear complexity. The strong performance on small/fast targets, which appear as sharp high-frequency streaks in spectrograms, further confirms MARSS’s capability in distinguishing meaningful signals from the substantial background clutter and speckle artifacts that dominate radar data.

We also compare our method with approaches on the CARRADA-RAC dataset. As shown in Table 2, the proposed model achieves superior performance on both the RA and Global views, while maintaining comparable results on the RD view.

Figure 7 illustrates segmentation under challenging cases (e.g., low SNR, small fast objects like the car) on CARRADA dataset. MARSS produces more accurate and temporally consistent masks than baselines.

4.3. Ablation Study

Our systematic ablation studies reveal how each module specifically addresses the radar phenomena described in our introduction. The baseline without proposed modules achieves 61.16% mIoU on RD and 41.04% on RA, high-

Table 2. Performance comparison on CARRADA-RAC. Results are reported in terms of mIoU and mDice for RD and RA views. Best results are **bold**, second-best are underlined.

Frameworks	#Params@Frames	RD View		RA View		Global View	
		mIoU	mDice	mIoU	mDice	mIoU	mDice
TMVA-Net	5.6M@5	59.7%	69.9%	46.6%	57.9%	53.2%	63.9%
PKCIn-Net	6.3M@5	60.6%	72.4%	47.3%	58.7%	54.0%	65.6%
AdaPKC ^θ -Net	6.3M@5	60.7%	72.7%	48.1%	59.6%	54.4%	66.2%
AdaPKC ^ξ -Net	6.3M@5	61.6%	73.6%	47.9%	59.3%	<u>54.8%</u>	66.5%
AdaPKC ^ξ -Net ^{FiT}	6.3M@5	60.8%	72.7%	<u>48.8%</u>	<u>60.5%</u>	<u>54.8%</u>	<u>66.6%</u>
MARSS(ours)	9.3M@5	<u>61.4%</u>	<u>73.4%</u>	49.0%	60.9%	55.2%	67.2%

Table 3. Ablation Study: Assess the contribution of each module (RADE, RFAF, RADM) in our framework. Results are reported in terms of mIoU and mDice for RD and RA views. Best results are **bold**, second-best are underlined.

RADE	RFAF	RADM	RD View		RA View	
			mIoU	mDice	mIoU	mDice
			61.16%	72.91%	41.04%	50.49%
✓			62.35%	74.24%	41.57%	51.69%
	✓		62.45%	74.48%	44.70%	55.69%
		✓	62.52%	74.35%	45.33%	<u>56.69%</u>
✓	✓		62.89%	<u>74.72%</u>	<u>45.33%</u>	<u>56.59%</u>
✓		✓	62.43%	74.09%	43.53%	54.09%
	✓	✓	<u>62.90%</u>	74.64%	43.76%	54.71%
✓	✓	✓	63.26%	75.18%	46.97%	58.78%

lighting the particular difficulty of RA view processing. The RADE module’s +1.19% mIoU improvement on RD view demonstrates its effectiveness in combating radar’s characteristic noise dominance through coordinated channel and spatial attention. Meanwhile, the RFAF module’s substantial +3.66% mIoU gain on RA view validates its multi-scale fusion capability for handling directional imbalance between range and angular information.

Most significantly, the RADM decoder achieves the strongest individual impact with +4.29% mIoU improvement on RA view, directly validating its anisotropic modeling capabilities through Mamba state space mechanisms. The synergistic performance when combining all three modules reveals their complementary nature: while RADE+RFAF and RFAF+RADM achieve similar RD performance (62.89-62.90% mIoU), the complete MARSS configuration reaches optimal 63.26% mIoU on RD and 46.97% on RA. The most telling synergy appears on RA view, where the full framework surpasses the best two-module combination by +1.64% mIoU, demonstrating that all three modules are essential for comprehensively addressing radar’s unique challenges, from noise suppression and multi-scale integration to directional dependency modeling.

5. Conclusion

We have presented MARSS, a modular attention-based architecture for radar semantic segmentation that explicitly tackles radar’s unique challenges. By embedding customized modules, RADE for noise-robust encoding, RFAF for multi-scale and regional feature selection, and RADM combining state space models for anisotropic decoding, MARSS effectively preserves directional and temporal information. Experiments on the CARRADA benchmark demonstrate state-of-the-art performance, significantly outperforming prior art (e.g., AdaPKC). Our ablations confirm that each component contributes complementarily, with the state space model introduction in the decoding phase particularly enhancing long-range dependency modeling.

Despite achieving state-of-the-art performance, MARSS has certain limitations: its multi-module design increases structural complexity, and its generalization capability across diverse radar hardware configurations and extreme weather conditions requires further validation. Future work will focus on model compression to enhance efficiency in real-time systems, exploring cross-dataset domain adaptation methods to improve generalization, and extending the framework to multi-sensor fusion paradigms for robust all-weather perception.

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